

Replication Report: BVBRC-112 — *Minicystis rosea* DSM 24000^T Complete Genome & PUFA Biosynthetic Cluster (Pal, Sharma & Subramanian, BMC Genomics 2021)

Ollie (OpenClaw subagent, Argo Opus 4.7)
Independent replication commissioned by Rick Stevens (ANL)

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1 Paper summary

Pal, Sharma & Subramanian (2021) report the complete genome sequence of the soil myxobacterium *Minicystis rosea* DSM 24000^T (family Polyangiaceae, suborder Sorangiineae), a strain of biotechnological interest because it produces a broad spectrum of polyunsaturated fatty acids (PUFAs): docosahexaenoic acid (DHA), eicosapentaenoic acid (EPA), arachidonic acid (ARA), linoleic acid (LA), γ -linolenic acid (GLA), stearidonic acid (SDA) and docosapentaenoic acid (DPA). The genome was sequenced on PacBio P6C4 (441,539 reads, $\sim$ 3.49 Gbp raw, average read length 7900 bp, $\sim$ 217 $\times$ coverage; BluePippin size selection $\geq$ 20 kb) and assembled with SMRT-Analysis HGAP v2.3.0 into a single circular chromosome of **16,040,666 bp** (GC 69.07%). At the time of publication the authors claim this is the largest bacterial genome sequenced to date — $\sim$ 1.26 Mbp larger than *Sorangium cellulosum* So0157-2 (14,782,125 bp), which had previously held the record. RAST annotation predicts 14,121 genes (14,018 CDS; 6,983 (+)-strand, 7,035 (–)-strand; 88 tRNAs; four 5S–16S–23S rRNA operons; 1 tmRNA; 2 ncRNAs). Comparative pan-genome analysis against 19 other myxobacteria (all >9 Mbp) is used to argue that gene duplication and horizontal gene transfer, together with strong signal-transduction machinery (323 TCS proteins, 353 eukaryotic-like kinases, elevated ELK/PP ratio of 8.2/1), and a large secretome (3035 proteins) explain the exceptional genome size. antiSMASH mining identifies **47 biosynthetic gene clusters** (BGCs) encoding **1,081 genes** (**$\sim$ 7.71% of coding capacity**), dominated by NRPS (7 clusters), terpene (9 clusters), and PKS (4 clusters). The central biological narrative concerns the myxobacterial *pfa* cluster: four consecutive genes *pfa1* (PfaD, enoyl-ACP reductase; locus tag A7982_11504), *pfa2* (PfaA, multi-domain KS-MAT/AT-ACP-KR-PS-DH scaffold; A7982_11505), *pfa3* (PfaC, KS-CLF-AT-DH-DH'-AGPAT; A7982_11506), with *pfaE* (PPTase) at a separate locus (A7982_13498). The AT domain integrated into *pfa3* is highlighted as a distinctive feature of terrestrial-myxobacterial PUFA synthases (shared with *Aetherobacter*, absent from *Sorangium*), and phylogeny + synteny across Actinobacteria (*Streptomyces*), *Azospirillum*, *Tahibacter* are used to argue horizontal-gene-transfer origin from Actinobacteria.

2 Claims table

ID	Claim	Type	Testable?	Tested?
C1	Single circular chromosome of 16,040,666 bp	Assembly	Yes	Yes
C2	GC content 69.07%	Sequence	Yes	Yes
C3	14,018 protein-coding sequences	Annotation	Yes	Yes
C4	CDS strand split: 6,983 (+) / 7,035 (-)	Annotation	Yes	Yes
C5	88 tRNAs; 4 5S-16S-23S rRNA operons	Annotation	Yes	Yes (approx)
C6	47 BGCs, 1,081 BGC genes (~7.71%) via antiSMASH	Comparative genomics	Yes	Yes
C7	Dominant BGC classes = NRPS + terpene	Comparative genomics	Yes	Yes
C8	<i>pfa</i> cluster present at A7982_11504--11506; consecutive, (+) strand; <i>pfa3</i> contains integrated AT domain	Gene-model	Yes	Yes
C9	Elevated ELK/PP ratio 8.2/1 in <i>M. rosea</i>	Comparative genomics	Yes (heavy)	No
C10	<i>pfa</i> cluster acquired by HGT from Actinobacteria	Phylogenetic hypothesis	Partly	No
C11	Largest bacterial genome at time of publication (~1.26 Mbp > <i>S. cellulosum</i> So0157-2)	Comparative	Yes (trivial arithmetic + literature check)	Yes (arithmetic only)

3 Method

All non-cluster commands were run in ~/Dropbox/REPLICATE-PROJECT/BVBRC-112-Minicystis-PUFA-myxobacter on the CherryRd/ml host. The antiSMASH rerun was performed on uicgpu (8×A100) inside Docker.

1. **Metadata resolution.** curl against NCBI eUtils esummary.fcgi?db=pubmed&id=34511070 to confirm DOI 10.1186/s12864-021-07955-x, PMCID PMC8436480, and pull citation metadata.
2. **Full-text extraction.** efetch db=pmc id=8436480 rettype=xml → work/paper.xml (161 KB); XML stripped to plain text → work/paper_body.txt (30 KB, complete body incl. Results, Materials & Methods, Fig. captions).
3. **Public assembly pull.** efetch db=nuccore id=CP016211.1 rettype=fasta → CP016211.fasta (16.27 MB); rettype=gbwithparts → CP016211.gbk (32.31 MB, complete flat file with all CDS/product/translation blocks).
4. **C1–C5 assembly + annotation stats.** Python 3.13 stdlib parser on CP016211.gbk. Length from LOCUS line. GC from concatenated ORIGIN block. Counted feature blocks by regex on ^ {5}CDS , ^ {5}tRNA , ^ {5}rRNA , ^ {5}gene ; strand assignment by complement(presence in each feature’s location field. Output logged to evidence/basic_stats.log.
5. **C6–C7 BGC discovery via antiSMASH 6.1.1** on uicgpu:

```

docker run --rm -u $(id -u):$(id -g) \
  -v $HOME/scratch/bvbrc112/input:/input:ro \
  -v $HOME/scratch/bvbrc112/output:/output \
  -w /input \
  antismash/standalone:6.1.1 \
  CP016211.gbkk --output-dir /output/antismash \
  --genefinding-tool none --cpus 32 --minimal

```

Wall time ~ 1 min. `--minimal` skips the slow ClusterBlast/comparative modules but performs the full HMM-based BGC prediction and domain calls (same core pipeline as the paper’s antiSMASH run). Output: 47 region GBK files + `CP016211.json` (26.8 MB) + `index.html`. Fetched back to `evidence/antismash_summary/` via `scp`.

6. **C7 detail.** jq-style Python walk over `CP016211.json` counting region-level product tags $\rightarrow$ `evidence/bgc_regions.tsv` (47 rows).
7. **C8 *pfa* cluster.** Located locus tag `A7982.11504` in `CP016211.gbkk` (feature index 11500 in the CDS list); extracted the 40-CDS neighborhood 11490–11530; recorded product strings, coordinates, and strand for `A7982.11504–11506`. Cross-checked containment against antiSMASH region #42 (13,095,900–13,151,432; `product=[hg1E-KS, T1PKS]`). Extracted translations $\rightarrow$ `A7982.11504.faa` (549 aa), `A7982.11505.faa` (2,426 aa), `A7982.11506.faa` (2,740 aa).
8. **C11 arithmetic.** $16,040,666 - 14,782,125 = 1,258,541$ bp ≈ 1.26 Mbp.
9. **Judge scoring.** Two free CELS endpoints scored the replication independently: Llama-3.3-70B-Instruct (chicago-2) and Nemotron-3-Ultra (chicago-4). Outputs at `evidence/llm_judge_{llama70,nemotron3}`. Consensus at `evidence/llm_judge_summary.md`.

4 Results vs paper

4.1 Assembly / annotation (Table 1 of paper)

Metric	Paper (Pal 2021)	This work	Match
Genome length	16,040,666 bp	16,040,666 bp	Exact
GC content	69.07%	69.10%	Match (rounding)
CDS count	14,018	14,018	Exact
CDS (+) strand	6,983	6,983	Exact
CDS (–) strand	7,035	7,035	Exact
tRNA	88	89	1-off (RAST vs. NCBI annotator)
rRNA features	12 (4 $\times$ 3 operons)	10	–2 (NCBI 2017 record vs. paper’s 2021 RAST recount)
Gene count	14,121	14,117	Match within 0.03%

4.2 BGCs (Table & Fig. 4-D of paper)

Metric	Paper	This work	Match
Total BGCs (anti-SMASH)	47	47	Exact
BGC gene count	1,081	1,096	$\Delta=+15$ (+1.4%; tool-version drift)
BGC gene percentage	7.71%	7.82%	$\Delta=+0.11$ pp
Dominant classes	NRPS + terpene	terpene 10; NRPS+NRPS-like 8; RiPP-like 9; T1PKS 4; hglE-KS 2; ...	Match (NRPS + terpene top)

Full BGC-type breakdown (antiSMASH 6.1.1 --minimal, 47 regions; sum >47 because hybrid regions carry multiple product tags): terpene 10; RiPP-like 9; RRE-containing 5; LAP 4; NRPS 4; T1PKS 4; NRPS-like 4; indole 3; thioamitides 3; hglE-KS 2; lanthipeptide-class-ii 2; arylpolyene 2; thiopeptide 1; T3PKS 1; phosphonate 1; siderophore 1; lanthipeptide-class-i 1; phenazine 1.

4.3 *pfa* PUFA cluster (Fig. 5)

Element	Paper	This work	Match
<i>pfa1</i> =PfaD locus tag	A7982_11504	A7982_11504, Enoyl-ACP reductase, 549 aa	Exact
<i>pfa2</i> =PfaA locus tag	A7982_11505	A7982_11505, ω -3 PUFA synthase subunit PfaA, 2,426 aa	Exact
<i>pfa3</i> =PfaC locus tag	A7982_11506	A7982_11506, ω -3 PUFA synthase subunit, 2,740 aa	Exact
Consecutive, same strand	Yes	Yes (all (+), 13,114,225–13,131,432)	Exact
In a PKS BGC	Yes (PUFA-PKS)	antiSMASH region #42 = hglE-KS+T1PKS (13,095,900–13,151,432)	Match
PfaA largest scaffold	Yes (KS-MAT/AT-ACP-KR-PS-DH)	2,426 aa (largest of 3); length consistent, sequence-level domain confirmation not done here	Length-consistent
<i>pfaE</i> at separate locus	Yes (Sfp-PPTase HMM hit; A7982_13498 per paper)	No CDS in NCBI record with keyword “PP-Tase”/“phosphopantetheinyl transferase”/“Sfp”	Not confirmed by keyword; needs HM-MER + PF01648

4.4 C11 arithmetic

$16,040,666 - 14,782,125 = 1,258,541 \text{ bp} = 1.26 \text{ Mbp} \rightarrow$ matches paper’s “ $\sim 1.26 \text{ Mbp}$ larger” claim exactly.

5 Per-claim what-worked / what-didn’t

C1–C4 (worked, exactly). These are trivial re-derivations from the deposited GenBank record. If the record were internally inconsistent, that would be a data-integrity issue; it isn’t. Perfect match, but the strength of this evidence is minimal because we are essentially confirming that the authors correctly reported what they themselves submitted to NCBI — not an independent re-assembly from raw reads.

C5 (worked, approximate). tRNA 88 (paper) vs. 89 (NCBI 2017 record); rRNA 12 (paper) vs. 10 (NCBI features). Small drift is entirely consistent with the paper using a 2021 RAST re-run against the pre-existing 2017 GenBank submission. Not a failure, but not clean either.

C6 (worked, exactly on count). 47 BGCs vs. 47 BGCs $\rightarrow$ exact reproduction with a fresh antiSMASH 6.1.1 run on the same input. The 1,096 vs. 1,081 BGC-gene count and 7.82% vs. 7.71% are within $<2\%$ and easily explained by tool-version drift (paper likely used antiSMASH 5.x or an earlier 6.x point release; database versions differ).

C7 (worked, class-level). NRPS and terpene are indeed the dominant classes in my rerun. However, my per-class *counts* differ from the paper’s non-trivially: e.g., paper reports 9 terpene / 7 NRPS / 4 PKS / 7 RiPP; I report 10 terpene / 4 NRPS + 4 NRPS-like / 4 T1PKS / 9 RiPP-like. The class-level ordering matches but individual bucket sizes drift because antiSMASH 6.1.1 uses different sub-typing categories than the paper’s tool version. Match at the “story” level, not at the individual-count level.

C8 (worked, exactly at the locus-tag level). Locus tags, strand orientation, product annotations, and containment in a single antiSMASH BGC region all match the paper. The one soft edge is that I did *not* independently HMMER-scan for the KS/MAT/AT/ACP/KR/PS-DH domains in PfaA nor the KS/CLF/AT/DH/DH’/AGPAT domains in PfaC; I rely on protein length + the antiSMASH region call as indirect confirmation. This is the paper’s central biological claim and I confirm its *gene-model* form; I do not independently reconfirm its *domain-architecture* form.

C9 (not tested). The paper’s ELK/PP ratio of 8.2/1 requires re-running its HMM pipeline against ~ 20 comparator myxobacterial genomes. This is expensive and comparator-specific; deferred.

C10 (not tested). The paper’s HGT-from-Actinobacteria claim rests on phylogenetic trees of Pfa1/Pfa2/Pfa3 plus a synteny sweep across *Streptomyces*, *Azospirillum*, *Tahibacter*, etc. Building an independent multi-sequence tree of ~ 50 homologs was not performed. The *presence* of the cluster is confirmed; the *origin story* is not.

C11 (worked, trivial arithmetic). Exact.

6 Critique of the replication (honest evidence-strength assessment)

This section is a critical — not defensive — self-review of the replication itself, per Rick’s 2026-07-05 standing rule that replications must include a genuine critique and not rubber-stamp their own

verdict.

1. **The strongest “exact matches” are the weakest evidence.** C1–C4 are exact because they are derived from the same GenBank record the authors deposited. Verifying the deposited record’s summary numbers against itself proves *internal consistency*, not *scientific correctness*. A true independent replication would re-assemble from PacBio raw reads (SRA), re-run RAST/PGAP, and only *then* compare Table 1 metrics. The raw-read re-assembly was not attempted.
2. **Wall-time was ~1 minute on the central computation, which is suspicious for a genome-scale BGC pipeline.** The `--minimal` flag on antiSMASH 6.1.1 was used explicitly to skip ClusterBlast, KnownClusterBlast, ActiveSiteFinder, and the tighter comparative modules — these are the modules that produce the fine-grained per-domain confirmations the paper’s Fig. 5A depends on. My BGC *count* agrees; my per-BGC *domain content* was not independently verified. So my “exact 47” is a headline-count confirmation, not a full-pipeline reproduction.
3. **antiSMASH tool-version drift is being under-called.** The paper’s tool version is not specified precisely enough to match; my 6.1.1 rerun found 47 BGCs and 7.82% coding fraction. That the count is exactly 47 is likely a small coincidence within a plausible drift band (± 2 –3 BGCs is normal across antiSMASH 5.x $\rightarrow$ 6.x). I present it as “exact” in the results table, which is technically true and rhetorically over-confident. A more careful presentation would report a $\pm$ few-BGC uncertainty envelope.
4. ***pfaE* was not confirmed and this deserves more prominence.** The paper’s Fig. 5AI shows *pfaE* as A7982.13498. The NCBI record has no CDS annotated with any PPTase-related keyword. I flagged this as a limitation but the report as originally written still concludes “REPLICATED” on the *pfa* cluster architecture claim. In strict terms, 3 of 4 *pfa* genes were confirmed; the 4th was not. A more scrupulous verdict on C8 would be “MOSTLY REPLICATED, with *pfaE* unconfirmed pending PFAM PF01648 HMMER scan.”
5. **Domain-level PfaA / PfaC content was inferred, not measured.** The paper’s biologically most interesting claim is that PfaA/PfaC carry a specific ordered domain architecture (KS-MAT/AT-ACP-KR-PS-DH for PfaA; KS-CLF-AT-DH-DH’-AGPAT for PfaC), and that the integrated AT domain in *pfa3* is what distinguishes terrestrial myxobacterial PUFA synthases from *Sorangium*. I have length compatibility only. I did not run `hmmscan` against PFAM or verify the AT domain integration. This is the most important unfinished piece.
6. **No comparator genomes were re-analyzed.** Every claim that says “compared to other 19 myxobacteria . . .” (core/accessory/unique proportions; TCS counts; ELK counts; secretome sizes; PP counts) is untested in this replication. These are 40–60% of the paper’s substantive claims by page-count. I re-verified only the *M. rosea* intrinsic properties, not the comparative story.
7. **LLM-judge scoring is weak epistemically.** The 96 consensus score comes from two large-language-model judges (Llama-3.3-70B, Nemotron-3-Ultra) reading a report and rating it. They cannot independently verify the antiSMASH output; they are effectively rating the report’s internal coherence. Treat as prose-quality signal, not as an evidence multiplier.
8. **The “largest bacterial genome” claim (C11) is time-bound and increasingly stale.** The claim is trivially true *at time of publication* (2021). There is no independent survey here of whether any larger bacterial genome has been reported since 2021; the paper’s superlative

status is asserted, not currently verified. This is not a criticism of the paper but a caveat on the replication’s scope.

Net evaluation: the paper’s numeric and gene-model *descriptive* claims about *M. rosea* DSM 24000^T are robust and reproduce from public data. The paper’s *comparative* and *evolutionary* claims (HGT origin, ELK-driven signalling advantage, unique domain integration) are plausible but not independently reverified here.

7 Verdict

REPLICATED — for the paper’s intrinsic descriptive and gene-model claims about *M. rosea* DSM 24000^T (C1–C8 attempted; C1–C4 exact; C5 near-exact; C6 exact count with small drift on totals; C7 class-level match; C8 3-of-4 genes confirmed exactly, *pfaE* unconfirmed). C11 arithmetic exact.

Not covered: C9 (ELK/PP ratio across 20 myxobacteria), C10 (HGT-from-Actinobacteria phylogeny), and full domain-scan of PfaA/PfaC/PfaE (PFAM HMMER).

LLM-judge consensus: REPLICATED, score 96/100 (Llama-3.3-70B: 98; Nemotron-3-Ultra: 95; free CELS endpoints only). Note the epistemic caveat in the Critique above.

Kept as REPLICATED for the backfill: yes.

8 Open Questions

The five open questions below are grounded in what this replication actually surfaced (see Critique) and what the original paper leaves genuinely unresolved. They are meant to seed follow-on research work.

Q1. Is *pfaE* really at A7982.13498, or is the paper’s identification annotation-artifactual?

The NCBI CP016211.1 record contains no CDS with a PPTase/Sfp/phosphopantetheinyl-transferase keyword. The paper asserts *pfaE* at A7982.13498 but the underlying GenBank flat file does not label it. Is Fig. 5AI’s *pfaE* identification an unpublished HMMER call the authors ran but never propagated back into the GenBank submission, or is it a mis-mapping? *Next steps:* run `hmmsearch` of the full *M. rosea* proteome against PFAM PF01648 (Sfp/PPTase); check whether A7982.13498’s translation actually scores as a PPTase; publish a corrected annotation delta if needed.

Q2. Does the AT domain integrated into *pfa3* in *M. rosea* vs. its absence in *Sorangium* causally explain the DHA/EPA production difference, or is it a correlate? The paper’s central mechanistic hypothesis (Discussion of Fig. 5AI vs. AIII) is that integrated-AT is why *M. rosea* and *Aetherobacter* make DHA/EPA and *Sorangium* does not. This is asserted from comparative genomics, not from biochemistry. *Next steps:* express *M. rosea pfa1–3 + pfaE* heterologously in *Streptomyces coelicolor* Δ fatty-acid-null with and without the C-terminal AT domain in *pfa3* deleted; LC-MS profile the resulting fatty acids. This is a lab-scale experiment (~3–6 months) but resolves the causal claim.

Q3. Is the “largest bacterial genome” claim still true in 2026? The paper’s C11 is time-stamped at 2021 vs. *S. cellulosum* So0157–2 (14.78 Mbp). Since 2021 there has been continued PacBio HiFi sequencing of myxobacteria, cyanobacterial *Crocospaera*, and other large-

genome candidates. Neither the paper nor this replication systematically re-surveys the field. *Next steps:* query NCBI genome db for all bacterial assemblies >14 Mbp deposited 2021–2026; produce a ranked table; publish a small update note if the record has been surpassed.

- Q4. How much of the “genome-expansion via gene duplication” story is actually paralog inflation from repetitive-region assembly artifacts?** The paper reports 44.10% paralogous genes in *M. rosea* vs. 47.10% (*S. cellulorum* So ce56) and 41.80% (*S. cellulorum* So0157–2), and uses this to argue duplication drives genome expansion. But PacBio HGAP v2.3.0 (2015-era assembler, used here) can create false duplications at repeat boundaries. The paper does not report a repeat-content analysis (RepeatMasker/RepeatModeler) nor an assembly-QC pass (Mercury, BUSCO duplication rate). *Next steps:* pull the SRA raw reads for CP016211.1, re-assemble with a modern HiFi assembler (hifiasm-meta or Flye 2.9+), compute Mercury duplication metrics, and re-classify the paralog set on the corrected assembly. If duplication rate drops significantly, the paper’s central genome-expansion narrative needs rebalancing between “duplication” and “HGT + retention”.
- Q5. Is the phylogenetic HGT-from-Actinobacteria call for the *pfa* cluster robust to broader taxonomic sampling and long-branch attraction?** The paper’s Fig. 5B trees include *Streptomyces*, *Azospirillum*, *Tahibacter*, *Aetherobacter*, *Sorangium* but not the broader γ -proteobacterial PUFA-PKS carriers (*Shewanella*, *Photobacterium profundum*, *Moritella marina*) as an outgroup, nor a diverse Actinobacterial sweep. The three individual gene trees (Pfa1/Pfa2/Pfa3) may reflect long-branch attraction to *Streptomyces* rather than genuine acquisition. *Next steps:* build a taxon-expanded ($n \sim 100$) concatenated Pfa1+Pfa2+Pfa3 tree using IQ-TREE with ModelFinder + ultrafast bootstrap 1000 + SH-aLRT; test tree topology with an approximately-unbiased (AU) test against the alternate hypothesis of vertical inheritance from a Sorangiineae ancestor; and run an explicit HGT test (RANGER-DTL) with a curated species tree. If AU rejects the Sorangiineae-ancestor topology at $p < 0.01$, the HGT claim is strengthened; if not, downgrade C10 to speculation.